

CS390R: Project 3

March 22, 2023

Please submit all screenshots in one pdf file with each clearly labeled (The corresponding question) for grading. Code should be handed in as separate py files with the specified names. The writeup for challenge three should be its own pdf file. All other files will be ignored.

1. For this challenge you will need to write a ropchain to open, read and write flag.txt to stdout. Doing this will earn you 60 pts. For the other 40 pts, list gadgets you've found from ROPGadget and explain what you will use them for. To find a syscall gadget, you may want to use the `-multibr` flag.
2. Explain the following. What is the exploitable code in this challenge? Why can we not naively input a ropchain? How can we bypass the check for a buffer overflow? (40 pts) For the last 60pts, create an exploit that spawns a shell.
3. Describe the GOT layout of the challenge using the `'got'` command in GDB. Describe when each entry in the GOT is first populated during the runtime of the program. (20 pts) Submit an exploit of this challenge that calls the win function. (40 pts) The remaining points are earned from your writeup of this challenge.